Laboratory 3 Report

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This report presents a comparative study of anomaly detection methods for network intrusion detection, applied to a dataset of network traffic logs containing normal and malicious connections. We evaluate four families of approaches: (i) One-Class SVM under novelty and outlier detection settings, (ii) autoencoder-based reconstruction error, (iii) dimensionality reduction (autoencoder bottleneck and PCA) combined with OC-SVM, and (iv) unsupervised clustering (K-Means and DBSCAN). Our analysis demonstrates that novelty detection — training exclusively on normal traffic — consistently outperforms outlier detection, and that non-linear feature extraction via autoencoder bottleneck embeddings, paired with a properly calibrated OC-SVM ($\nu = 0.1$), achieves the best overall test performance (F1-Macro = 0.7795). We further show that small anomaly contamination during training (10–20%) is more damaging than full contamination, and that DBSCAN's density-based noise detection is fundamentally misaligned with anomaly detection in network traffic.

1 INTRODUCTION

Intrusion Detection Systems (IDS) are a critical component of modern network security, tasked with distinguishing legitimate traffic from malicious activity such as denial-of-service attacks, network probing, and unauthorized access attempts. A central challenge in this domain is that labeled anomaly data is often scarce or unavailable, motivating the use of anomaly detection methods that learn normality from benign traffic alone.

This laboratory investigates supervised, semi-supervised, and unsupervised anomaly detection techniques applied to a network traffic dataset derived from the KDD Cup 1999 benchmark. The dataset contains 43 features describing individual network connections, with labels indicating the traffic type (Normal, DoS, Probe, R2L).
Our analysis is structured around four main tasks:

(1) **Dataset characterization and preprocessing** (Section 2): We explore the dataset structure, handle categorical and numerical features, and analyze per-class feature distributions through heatmaps and correlation analysis.
(2) **Shallow anomaly detection** (Section 3): We evaluate One-Class SVM under three training configurations — normal data only (novelty detection), all data (outlier detection), and mixed contamination levels — quantifying the impact of the ν parameter and training data composition.
(3) **Deep anomaly detection** (Section 4): We train an autoencoder on normal data and explore three detection strategies: reconstruction error thresholding, OC-SVM on bottleneck embeddings, and OC-SVM on PCA-reduced features.
(4) **Unsupervised clustering** (Section 5): We apply K-Means and DBSCAN to assess whether purely unsupervised approaches can identify attack patterns, using silhouette analysis, t-SNE visualization, and cluster purity metrics.

All experiments are conducted in Google Colab using scikit-learn and PyTorch, with an 80/20 stratified split performed before preprocessing to prevent data leakage.

2 DATASET CHARACTERIZATION AND PREPROCESSING

This section describes the dataset, the preprocessing pipeline, and the exploratory analysis performed to understand the feature space and identify discriminative patterns.

2.1 Dataset Exploration

The dataset consists of network traffic logs derived from the KDD Cup 1999 dataset, provided as separate training and test CSV files. Each record represents a network connection, described by 43 columns: 41 features, a multi-class attack label, and a binary label (`normal` vs. `anomaly`).

Feature types. The 41 features include 3 categorical attributes (`protocol_type`, `service`, `flag`) and 38 numerical attributes. The categorical features describe the connection protocol, the destination service, and the TCP connection status flag, respectively.

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Label distribution. The training set contains 18,831 samples with approximately 71.4% normal traffic and 28.6% anomalies. After an 80/20 stratified split, the training partition contains 15,063 samples and the validation partition contains 3,767 samples. The test set contains 5,826 samples with a notably different distribution: only 36.9% normal vs. 63.1% anomalous traffic. This distribution shift between training and test sets is a deliberate challenge that tests model generalization.

Class	**Train (80%)**	**Val (20%)**	**Test**
Normal	10,758 (71.4%)	2,690 (71.4%)	2,152 (36.9%)
DoS	2,330 (15.5%)	583 (15.5%)	2,634 (45.2%)
Probe	1,830 (12.1%)	458 (12.2%)	1,040 (17.9%)
R2L	145 (1.0%)	36 (1.0%)	0 (0.0%)
Total	15,063	3,767	5,826

Table 1. Label distribution across dataset splits.

The test set's anomaly-dominant composition — particularly the absence of R2L attacks and the surge in DoS (from 15.5% to 45.2%) — forces models to generalize beyond the training distribution.

2.2 Preprocessing Pipeline

The preprocessing pipeline is designed to ensure clean, standardized input for all downstream models while preventing data leakage.

Data cleaning. We first checked for missing values (`NaN`, `Inf`) and duplicates, finding one duplicate row which was removed. We then identified and dropped four zero-variance columns (`land`, `urgent`, `num_outbound_cmds`, `is_host_login`), as they provide no discriminative information. This reduced the feature space from 41 to 37 attributes.

Train/validation split. Before any feature transformation, we performed an 80/20 stratified split of the training data. This is a deliberate design decision: by splitting *before* preprocessing, we ensure that the scaler and encoder are fitted exclusively on training data, preventing any information leakage from the validation into the training pipeline.

Categorical encoding. For each categorical feature, we grouped rare values (below a frequency threshold) into an "other" category to reduce dimensionality and prevent overfitting to infrequent values. Specifically, we retained the 8 most frequent services, 5 flags, and 3 protocols. We then applied One-Hot Encoding with `handle_unknown='ignore'` to gracefully handle any unseen categories in the test set.

Numerical standardization. All numerical features were standardized using `StandardScaler`, fitted exclusively on the training partition. The same transformation was then applied to the validation and test sets.

After preprocessing, the final feature space consists of **51 dimensions**: 34 numerical features and 17 one-hot encoded categorical features.

2.3 Domain Analysis: Feature Heatmaps

To understand which features characterize each attack type, we computed per-class heatmaps of the mean, standard deviation, and median feature values (after standardization).

Key discriminative patterns.

- **DoS attacks** exhibit extremely elevated `serror_rate` and `srv_serror_rate` ($\mu > 2.0$ standard deviations above the global mean), reflecting SYN flood behavior. They also show high `count` and `srv_count` values, consistent with connection floods.
- **Probe attacks** are characterized by high `dst_host_srv_diff_host_rate` and elevated service diversity metrics, reflecting port/host scanning patterns.
- **R2L attacks** show subtle signatures: slightly elevated `hot` and `is_guest_login` features, but otherwise closely resemble normal traffic — explaining why they are the hardest to detect.
- **Normal traffic** has low, stable values across all anomaly-indicative features, with moderate connection rates.

2.4 Feature Correlation Analysis

We computed the Pearson correlation matrix across all numerical features and identified several highly correlated pairs ($r > 0.95$), notably `serror_rate`/`srv_serror_rate` ($r = 0.981$) and `rerror_rate`/`srv_rerror_rate` ($r = 0.991$). These strong correlations suggest significant redundancy in the feature space, which motivates the dimensionality reduction methods explored in Section 4 (PCA and autoencoder bottleneck).

3 SHALLOW ANOMALY DETECTION

In this section, we investigate One-Class Support Vector Machine (OC-SVM) for anomaly detection under three training configurations: normal data only, all data, and mixed data with varying anomaly ratios.

3.1 OC-SVM with Normal Data Only (Novelty Detection)

In novelty detection, the OC-SVM is trained exclusively on normal traffic, learning a tight boundary around the normal data manifold. The key hyperparameter is ν, which controls the upper bound on the fraction of training errors.

ν parameter analysis. Since the training data is purely normal, ν should reflect the expected fraction of noisy or edge-case normal samples. We evaluated four values: $\nu \in \{0.001, 0.01, 0.05, 0.5\}$.

ν	**Train F1-Macro**	**Test F1-Macro**
0.001	0.9058	0.7382
0.01	0.9039	0.7401
0.05	**0.9026**	**0.7477**
0.5 (default)	0.6378	0.4647

Table 2. OC-SVM (Normal-Only): impact of ν on performance.

We selected $\nu = 0.05$ as the best estimate, as it achieves the highest test F1-Macro (0.7477). This value allows the model to exclude approximately 5% of normal samples as noise during training, creating a slightly looser boundary that generalizes better. The default $\nu = 0.5$ is catastrophic for novelty detection, as it forces half of the normal training data to be treated as outliers, effectively collapsing the decision boundary.

3.2 OC-SVM with All Data (Outlier Detection)

When training on the full dataset (normal + anomalies), the OC-SVM operates as an *outlier detector*. We estimated ν as the true contamination rate: $\nu = 4305/15063 \approx 0.286$.

Model	**Test F1-Macro**	**Test Accuracy**
Normal-Only (ν=0.05)	**0.7477**	0.7667
All-Data (ν=0.286)	0.6484	0.6656

Table 3. Comparison of Normal-Only vs. All-Data OC-SVM on the test set.

The Normal-Only model outperforms the All-Data model by approximately 10 percentage points in F1-Macro. This demonstrates that novelty detection (learning pure normality) is more effective than outlier detection (learning on contaminated data): the All-Data model's boundary is distorted by the inclusion of anomalies during training.

3.3 Sensitivity Analysis: Impact of Anomaly Contamination

To quantify the effect of training data contamination, we trained OC-SVMs with increasing proportions of anomalous data: 0%, 10%, 20%, 50%, and 100% of available anomalies. For each configuration, ν was set proportionally to the anomaly ratio.

Anomaly %	ν	Train F1-Macro	Test F1-Macro
0% (Normal-Only)	0.01	0.9039	**0.7401**
10%	0.038	0.6628	0.5137
20%	0.074	0.6779	0.5589
50%	0.167	0.7226	0.5975
100% (All-Data)	0.286	0.7364	0.6482

Table 4. Sensitivity analysis: impact of anomaly contamination on OC-SVM performance.

The results reveal a non-monotonic pattern. Small contamination levels (10–20%) cause the most damage: the model receives mixed signals about what constitutes "normal" behavior, but ν is too small to properly account for the anomalies. As contamination increases to 50–100%, performance partially recovers because ν grows large enough to capture the true anomaly proportion. However, no contaminated model matches the purely normal training baseline.

3.4 Robustness Evaluation

We compared three representative models on the test set to assess robustness:

Model	Test F1-Macro	Most Missed Attack
Normal-Only (ν=0.05)	**0.7477**	DoS (609 missed)
All-Data (ν=0.286)	0.6484	DoS (1,074 missed)
10% Anomalies	0.5137	DoS (1,989 missed)

Table 5. Robustness evaluation on the test set: three OC-SVM configurations.

Per-attack confusion analysis. DoS attacks generate the most false negatives across all models, both in absolute count and due to the distribution shift (DoS increased from 15.5% to 45.2% in the test set). R2L attacks are the hardest to detect proportionally, as their features closely mimic normal traffic patterns. The Normal-Only model shows the best overall balance between precision (fewer false alarms) and recall (catching more attacks).

4 DEEP ANOMALY DETECTION AND DATA REPRESENTATION

This section explores deep learning and dimensionality reduction approaches for anomaly detection: autoencoder-based reconstruction error, autoencoder bottleneck embeddings with OC-SVM, and PCA-based dimensionality reduction with OC-SVM.

4.1 Autoencoder Architecture and Training

We designed a symmetric autoencoder with the following architecture:

- **Encoder:** Input (51) $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 16 (bottleneck), with BatchNorm, ReLU, and Dropout (0.2) after each hidden layer.
- **Decoder:** 16 $\rightarrow$ 64 $\rightarrow$ 128 $\rightarrow$ 51 (reconstruction), with a symmetric architecture.

Composite loss function. Since the feature space contains both numerical and categorical (one-hot encoded) features, we used a composite loss: MSE for numerical features and Cross-Entropy for each one-hot encoded categorical group. This ensures the autoencoder learns appropriate reconstruction for each feature type.

Training procedure. The autoencoder was trained exclusively on *normal* validation data (10,758 samples) in a semi-supervised fashion: by learning to reconstruct only normal traffic, anomalies should produce higher reconstruction errors due to unfamiliar patterns.

4.2 Hyperparameter Tuning

We performed a grid search over learning rates $\{5 \times 10^{-3}, 10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$ with a maximum of 150 epochs and early stopping (patience = 15).

Learning Rate	Min Val Loss	Best Epoch	Early Stop?
5×10^{-3}	0.2160	26	Yes (epoch 41)
1×10^{-3}	0.1232	78	Yes (epoch 93)
$\mathbf{5 \times 10^{-4}}$	**0.1140**	**82**	Yes (epoch 97)
1×10^{-4}	0.1609	148	No

Table 6. Learning rate grid search results (bottleneck dimension = 16).

The learning rate 5×10^{-4} achieves the lowest validation loss (0.1140) at epoch 82 before early stopping. Higher learning rates converge faster but to worse optima, while 10^{-4} underfits within 150 epochs.

4.3 Reconstruction Error Analysis

Threshold selection. We computed the ECDF (Empirical Cumulative Distribution Function) of reconstruction errors on the autoencoder validation set (normal data only). The **95th percentile** was selected as the anomaly threshold: $\theta = 0.3604$. This means approximately 5% of normal data is expected to be misclassified as anomalous — an acceptable false positive rate that accounts for natural variation in normal traffic.

Anomaly detection performance. Samples with reconstruction error $> \theta$ are classified as anomalies.

Dataset	F1-Macro	Precision (Anomaly)	Recall (Anomaly)
Train Set	0.9132	0.8823	0.8690
Validation Set	0.9024	0.8713	0.8487
Test Set	**0.7547**	0.7994	0.8669

Table 7. Autoencoder reconstruction error anomaly detection ($\theta = 0.3604$).

The test F1-Macro of 0.7547 demonstrates strong generalization. The ECDF comparison across datasets shows that reconstruction errors are progressively higher on the full training set (which includes anomalies) and test set, confirming that the autoencoder successfully learned the normal data manifold: unseen anomalies produce significantly higher errors.

4.4 Latent Space Anomaly Detection (AE + OC-SVM)

We extracted the 16-dimensional bottleneck embeddings from the trained encoder and used them as input features for an OC-SVM trained on normal data only.

ν calibration. We tested two values: $\nu = 0.01$ (conservative) and $\nu = 0.1$ (accounting for the approximately 10% of normal data with elevated reconstruction error in the bottleneck space).

ν	Train F1-Macro	Test F1-Macro
0.01	0.7662	0.6645
0.1	**0.8300**	**0.7795**

Table 8. AE Bottleneck + OC-SVM: impact of ν on performance.

The $\nu = 0.1$ configuration yields a substantial improvement (+11.5 percentage points on the test set), making the AE Bottleneck + OC-SVM the **best-performing method overall** (test F1-Macro = 0.7795). The conservative $\nu = 0.01$ was too restrictive in the compressed 16-dimensional space, where normal points are more dispersed than in the original feature space.

4.5 PCA-based Anomaly Detection

As a linear baseline for dimensionality reduction, we applied PCA on normal training data and selected 20 components, capturing 95.5% of the cumulative explained variance. An OC-SVM ($\nu = 0.01$) was then trained on the PCA-transformed normal data.

Dataset	F1-Macro	Precision (Anomaly)	Recall (Anomaly)
Train Set	0.8835	0.9655	0.7206
Test Set	**0.7377**	0.8533	0.7186

Table 9. PCA + OC-SVM performance (20 components, 95.5% variance).

PCA + OC-SVM achieves F1-Macro = 0.7377 on the test set, nearly matching the original OC-SVM (0.7477) despite a 60% dimensionality reduction (51 → 20). This confirms significant redundancy in the feature space.

4.6 Comprehensive Method Comparison

Table 10 consolidates all anomaly detection methods from Tasks 2 and 3.

Task	Method	Dim.	Test F1-Macro
3	AE Bottleneck + OC-SVM (ν=0.1)	16	**0.7795**
3	AE Reconstruction Error	51	0.7547
2	OC-SVM Normal-Only (ν=0.05)	51	0.7477
3	PCA + OC-SVM	20	0.7377
2	OC-SVM All-Data (ν=0.286)	51	0.6484

Table 10. Comprehensive comparison of anomaly detection methods (test set).

The AE Bottleneck + OC-SVM achieves the highest test F1-Macro, demonstrating that the autoencoder's non-linear compression captures discriminative patterns that both linear PCA and the original feature space miss. The key factor is pairing the bottleneck representation with an appropriately calibrated ν parameter.

5 UNSUPERVISED ANOMALY DETECTION AND INTERPRETATION

This section explores purely unsupervised clustering approaches — K-Means and DBSCAN — for anomaly detection, evaluating their ability to separate attack types without label supervision.

5.1 K-Means Clustering

We applied K-Means with $k = 4$ clusters on the full training data, matching the number of known traffic types (Normal, DoS, Probe, R2L).

Cluster	Size	Silhouette	Dominant Class	Purity	Key Content
0	1,560	0.261	probe (54%)	Mixed	probe, normal, dos
1	1,422	0.753	dos (100%)	Pure	Nearly all DoS
2	9	0.392	normal (100%)	Pure	Normal outliers
3	12,072	0.461	normal (85%)	Mixed	normal, probe, dos, r2l

Table 11. K-Means cluster composition and quality ($k = 4$, training set).

Cluster analysis. The average silhouette score is 0.468, indicating moderate cluster separation. Cluster 1 achieves excellent separation (silhouette = 0.753) because DoS floods create distinctly different network traffic patterns (high SYN error rates, high connection counts). Cluster 0 has the lowest silhouette (0.261), corresponding to a boundary region mixing probe, normal, and some DoS traffic. All 144 R2L attacks are absorbed into Cluster 3 alongside normal traffic, confirming that R2L attacks are indistinguishable from normal behavior using Euclidean distance.

Post-hoc anomaly detection. Using majority voting to assign binary labels to clusters, K-Means achieves a training F1-Macro of 0.7954. However, this requires label access for evaluation and does not generalize to unseen data.

5.2 t-SNE Visualization

We applied t-SNE to project the training data into 2D, testing three perplexity values: 5 (local focus), 30 (balanced), and 100 (global). Perplexity = 30 provides the best balance between local and global structure.

Cluster vs. true label comparison. The side-by-side t-SNE comparison between K-Means cluster assignments and true attack labels reveals key insights: K-Means correctly separates the DoS cluster (Cluster 1 maps cleanly to the DoS group in t-SNE space), but Cluster 3 absorbs diverse traffic types that t-SNE shows as separable groups. Notably, R2L attacks completely overlap with normal traffic in the t-SNE projection, confirming the difficulty of their detection.

5.3 DBSCAN Clustering

DBSCAN identifies dense clusters and labels remaining points as noise (-1), which we interpret as potential anomalies. We investigated two configurations using cosine distance.

Parameter selection. The `min_pts` parameter was derived from K-Means analysis: the smallest pure-normal cluster across $k \in \{4, \ldots, 10\}$ had 7 points, yielding `min_pts` = 7. The ϵ parameter was selected from the k-distance graph.

Configuration comparison. We tested multiple `min_pts` values to understand the sensitivity:

min_pts	ϵ	Clusters	Noise %	F1-Macro	Noise: Normal	Noise: Anomaly
7	0.064	40	3.2%	0.4285	80.9%	19.1%
100	0.053	17	21.8%	0.4707	76.0%	24.0%
500	0.260	4	14.2%	0.4874	70.7%	29.3%
1600	0.635	1	2.0%	0.4466	52.5%	47.5%

Table 12. DBSCAN sensitivity analysis: impact of `min_pts` (cosine distance).

Analysis. No configuration yields effective anomaly detection (all F1-Macro < 0.5). With small `min_pts` (7), the noise cluster is dominated by normal outliers (80.9%). Increasing `min_pts` shifts the noise composition toward anomalies (47.5% anomalous at `min_pts` = 1600), but at the cost of collapsing all data into a single cluster. The `min_pts` = 500 configuration offers the most interpretable result with 4 clusters and 14.2% noise, but performance remains poor.

Why DBSCAN noise ≠ anomaly detection. The fundamental mismatch lies in the density assumption:

- **Dense attacks form valid clusters:** DoS floods generate thousands of repetitive connections that DBSCAN groups into dense clusters, not noise.
- **Sparse normal outliers trigger noise:** Normal traffic with unusual feature combinations (rare services, atypical durations) is isolated and flagged as noise.
- **R2L attacks hide in normal clusters:** R2L mimics legitimate behavior and remains embedded within normal-dominated clusters.

This demonstrates that density-based clustering alone cannot serve as a reliable intrusion detection system — the density structure of network traffic does not align with the normal/anomalous dichotomy.

6 CONCLUSIONS

This laboratory provided a comprehensive evaluation of anomaly detection methods for network intrusion detection. Our key findings are summarized below.

Novelty detection outperforms outlier detection. Training OC-SVM exclusively on normal traffic ($\nu = 0.05$, F1 = 0.7477) consistently outperforms training on contaminated data ($\nu = 0.286$, F1 = 0.6484). Small contamination levels (10–20%) are paradoxically the most damaging, as they distort the decision boundary without providing sufficient anomaly signal.

Non-linear feature extraction is beneficial. The autoencoder bottleneck + OC-SVM ($\nu = 0.1$) achieves the best overall test F1-Macro of **0.7795**, demonstrating that the autoencoder's non-linear compression captures patterns that both linear PCA and the original feature space miss. However, performance is highly sensitive to the ν parameter: $\nu = 0.01$ in the same bottleneck space yields only F1 = 0.6645.

Reconstruction error is a strong stand-alone method. Autoencoder reconstruction error with a 95th-percentile threshold achieves F1 = 0.7547 without requiring any additional classifier, making it the simplest competitive approach.

Unsupervised clustering has fundamental limitations for IDS. K-Means correctly isolates DoS attacks into a pure cluster but absorbs all R2L attacks into the normal-dominated cluster. DBSCAN's noise detection is fundamentally misaligned with anomaly detection: dense attack patterns form valid clusters, while sparse normal outliers are flagged as noise. Neither clustering approach is viable as a standalone intrusion detection system.

Distribution shift remains a challenge. The test set's drastically different composition (63% anomalies vs. 29% in training) causes significant performance drops across all methods, highlighting the importance of models that learn robust representations of normality rather than memorizing specific attack patterns.